Horticulture 1 · Unit 5 · Level 2 (modified)

Soil & Nutrients



TODAY'S PLAN

- 🌱 Growing media
- 🪨 Soil texture
- 🫙 Jar test
- 🟢 N-P-K

Review · 5 minutes

Quick Review



🔎 Name 2 trailing bedding plants.

🏷️ Name 3 fields on the Plant ID card.

🌸 Which bedding plants have composite flowers?

Words to Know

Words to Know



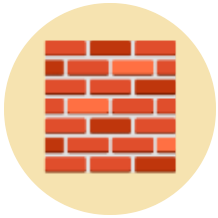
Sand

The largest soil particles. Water drains fast through sand.



Silt

Medium-sized particles that feel smooth like flour.



Clay

The smallest particles. Clay is sticky and holds water.



Loam

A mix of sand, silt and clay that holds water and drains well.

Words to Know

More Words to Know



🪴 **Potting mix**: A growing medium (peat, bark, perlite) used in greenhouse pots instead of garden soil.

🟢 **Nitrogen (N)**: Nutrient for green, leafy growth. Too little = yellow older leaves.

🟣 **Phosphorus (P)**: Nutrient for roots, flowers and seeds. Too little = purple leaves, poor roots.

🟠 **Potassium (K)**: Nutrient for strong stems and disease resistance. Too little = brown, scorched leaf edges.

Learn

Plants Need



Light
for photosynthesis



Water
not soggy



Temperature
warm, not hot



Media + nutrients
soil and fertilizer

Learn

Growing Media



Greenhouses use **potting mix** (peat, bark, **perlite**), not garden soil.

It drains well and has no weed seeds.

Wet dry mix first; dust can carry germs, so no hand-to-mouth.

Learn

Soil Particles



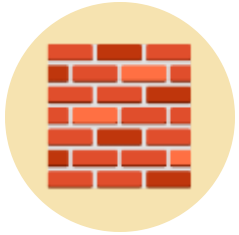
Sand

largest, drains fast



Silt

medium, smooth



Clay

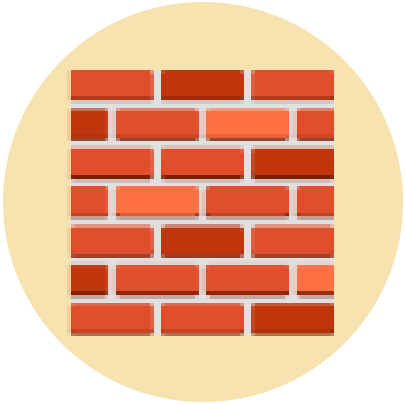
smallest, sticky, holds water



Loam

balanced mix

🃏 Check 1 of 3



Which soil holds the most water?

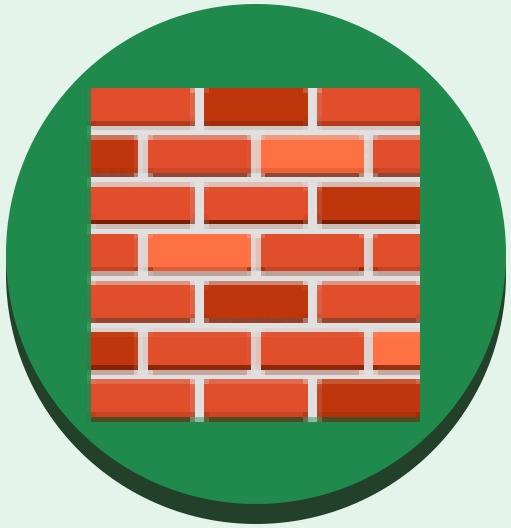
Hold up 1, 2, or 3:

1️⃣ Sand

2️⃣ Clay

3️⃣ Gravel

✅ Answer · Card 1 of 3



Clay

Tiny particles hold water.

Steps

Soil Jar Test



1. Put 2 inches of soil in a jar; record the depth
2. Add water, lid on, shake 10 seconds
3. Let it settle overnight
4. Measure each layer: sand (bottom), silt, clay (top)
5. Percent = layer ÷ total × 100

🃏 Check 2 of 3



In the jar, which layer settles at the bottom?

Hold up 1, 2, or 3:

1️⃣ Clay

2️⃣ Silt

3️⃣ Sand

✅ Answer · Card 2 of 3



Sand

Heaviest first.

Learn

N-P-K

N Nitrogen
green leafy growth

P Phosphorus
roots, flowers, seeds

K Potassium
strong stems, disease resistance

Bag numbers
10-10-10 = % N-P-K

Learn

Deficiency Signs



Low N

yellow older leaves



Low P

purple leaves, poor roots



Low K

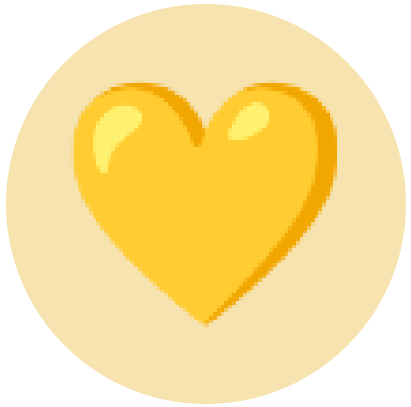
brown scorched edges



Too much

burned roots, weak floppy growth

🃏 Check 3 of 3



Older leaves turning yellow suggests low…

Hold up 1, 2, or 3:

1️⃣ Nitrogen

2️⃣ Potassium

3️⃣ Water only

✅ Answer · Card 3 of 3



Nitrogen

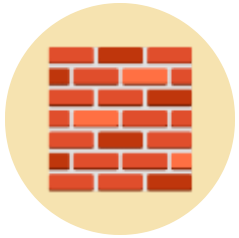
N moves to new leaves.

Wrap-Up

Recap



Potting mix
drains, clean



Clay
holds water



Jar test
layer %



N-P-K
leaves, roots, strength

Wrap-Up

Exit Ticket



What does each letter of N-P-K do?

Which soil particle is smallest?

How do you find the percent of sand in the jar?